Skill Builder Pack 1 - Importing Data into Excel

Purpose

In many workplaces, data is stored in different systems and exported into files such as CSV downloads, Excel spreadsheets, or PDF reports.

Before any analysis can take place, the data must first be **imported into Excel correctly**.

This skill builder focuses on **learning how to bring data into Excel from different sources**.

You will practise importing data from:

- CSV files
- Excel workbooks
- PDF documents

These are very common file types used when working with data.

Learning Outcomes

By the end of this skill builder you should be able to:

- Import a **CSV file** into Excel
- Import data from another **Excel workbook**
- Import a **table from a PDF document**
- Use Excel's **Get Data tools**
- Load imported data into Excel tables

Scenario

You are working in a **Student Engagement Team** at a university.

The team monitors student attendance and engagement to help identify students who may need additional support.

Data is exported from several systems into different file types.

Your job is to **bring the data into Excel so it can later be analysed**.

Files Provided

You will receive the following files separately:

- attendance_march.csv
- student_details.xlsx
- engagement_scores.csv
- withdrawals_report.pdf

Each file contains different information about students.

Skill 1 – Importing a CSV File

CSV stands for **Comma Separated Values**.

It is one of the most common ways that systems export data.

A CSV file contains rows of data separated by commas.

Example:

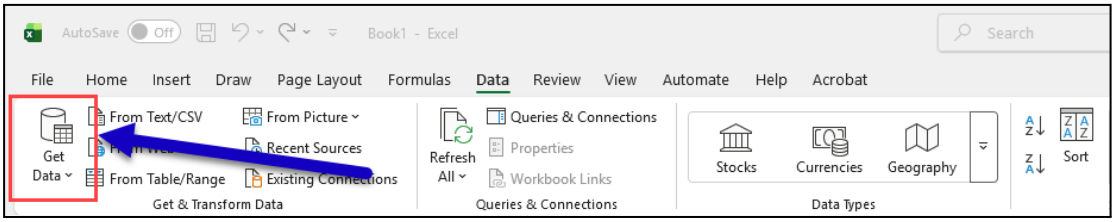
```
student_id,student_name,module,attendance_status
1001,Ali Khan,Data Fundamentals,Present
```

Step-by-Step

1. Open a new Excel workbook

2. Select the **Data** tab

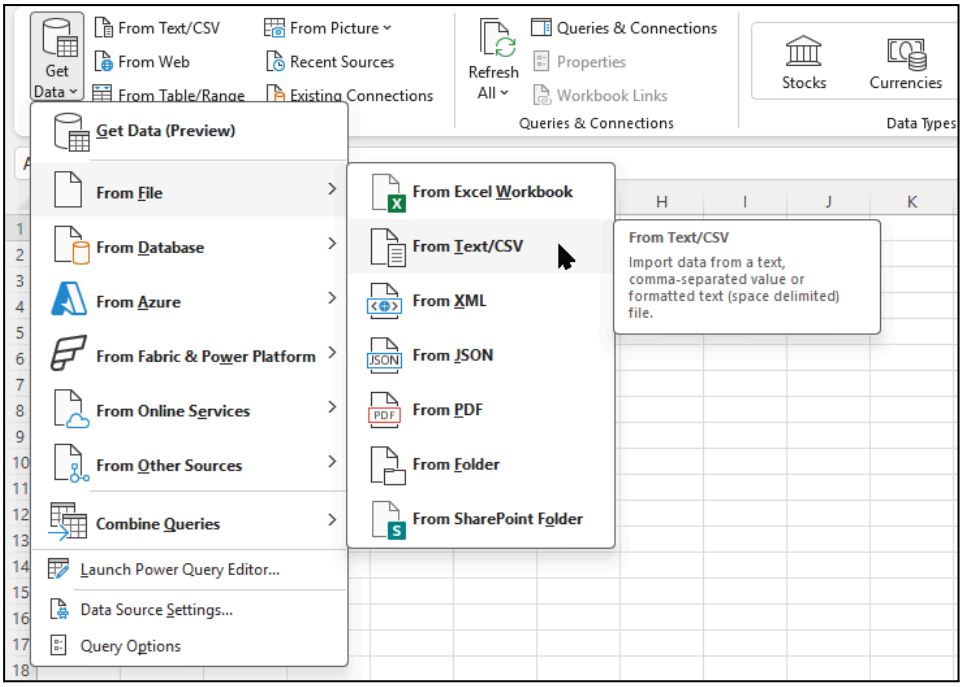
3. Click **Get Data**



Why Get Data?

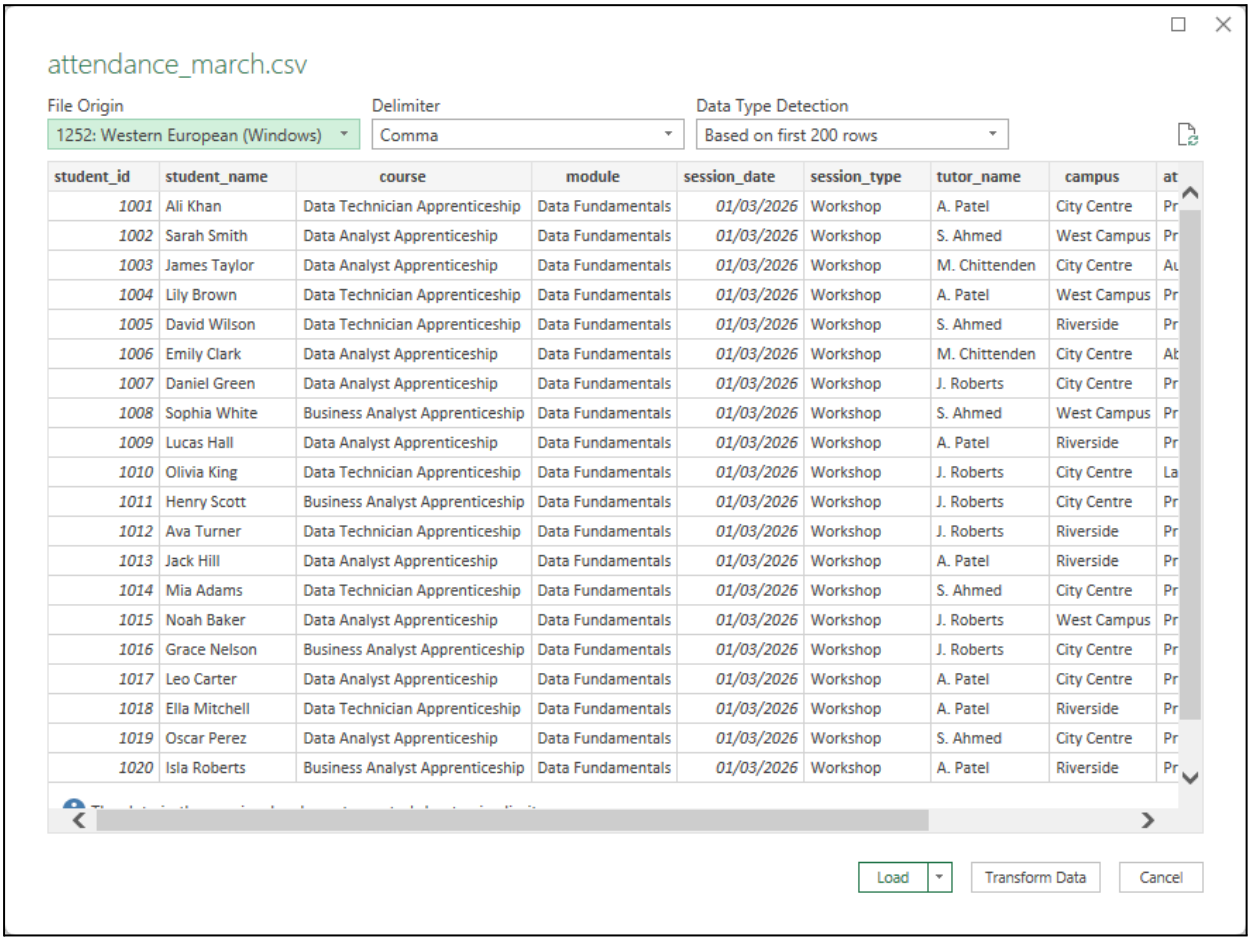
When working with data in Excel, it is better to use **Data → Get Data** rather than simply opening a file. Opening a file (such as a CSV or another spreadsheet) only loads the data into Excel as a static table, which means Excel treats it as a one-off copy. If the original data changes, the process must be repeated manually. Using **Get Data**, however, imports the file through Excel's data connection tools (Power Query). This allows Excel to remember where the data came from and how it was imported. As a result, the data can be refreshed automatically if the source file is updated, and any preparation steps—such as formatting columns or removing errors—can be applied consistently each time the data is loaded. For this reason, **Get Data is the preferred method when importing data for analysis**, as it creates a repeatable and more reliable process.

4. Choose **From File → From Text/CSV**



5. Select the file **attendance_march.csv**

Excel will show a **preview window**.

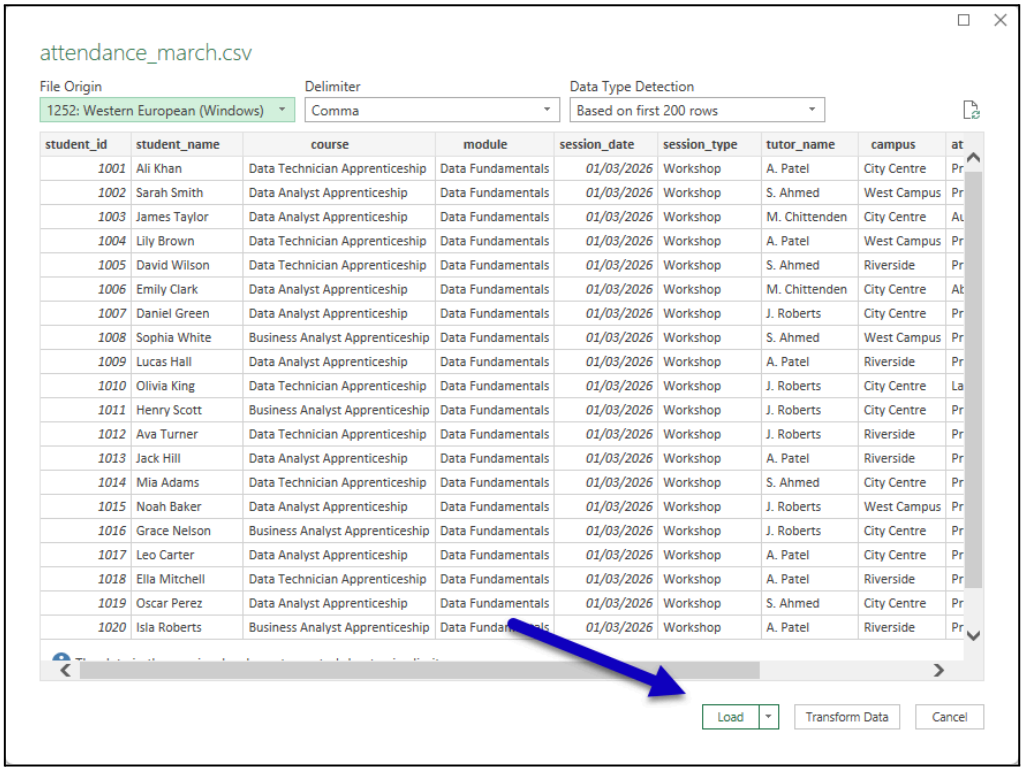


student_id	student_name	course	module	session_date	session_type	tutor_name	campus	at
1001	Ali Khan	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	City Centre	Pr
1002	Sarah Smith	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Pr
1003	James Taylor	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	M. Chittenden	City Centre	Au
1004	Lily Brown	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	West Campus	Pr
1005	David Wilson	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	Riverside	Pr
1006	Emily Clark	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	M. Chittenden	City Centre	Ab
1007	Daniel Green	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Pr
1008	Sophia White	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Pr
1009	Lucas Hall	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Pr
1010	Olivia King	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	La
1011	Henry Scott	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Pr
1012	Ava Turner	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	Riverside	Pr
1013	Jack Hill	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Pr
1014	Mia Adams	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	City Centre	Pr
1015	Noah Baker	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	West Campus	Pr
1016	Grace Nelson	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Pr
1017	Leo Carter	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	City Centre	Pr
1018	Ella Mitchell	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Pr
1019	Oscar Perez	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	City Centre	Pr
1020	Isla Roberts	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Pr

This allows you to check that:

- The columns are separated correctly
- The data appears structured

Click **Load** to import the data.



student_id	student_name	course	module	session_date	session_type	tutor_name	campus	at
1001	Ali Khan	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	City Centre	Pr
1002	Sarah Smith	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Pr
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1004	Lily Brown	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	West Campus	Pr
1005	David Wilson	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	Riverside	Pr
1006	Emily Clark	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	M. Chittenden	City Centre	Ab
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1012	Ava Turner	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	Riverside	Pr
1013	Jack Hill	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Pr
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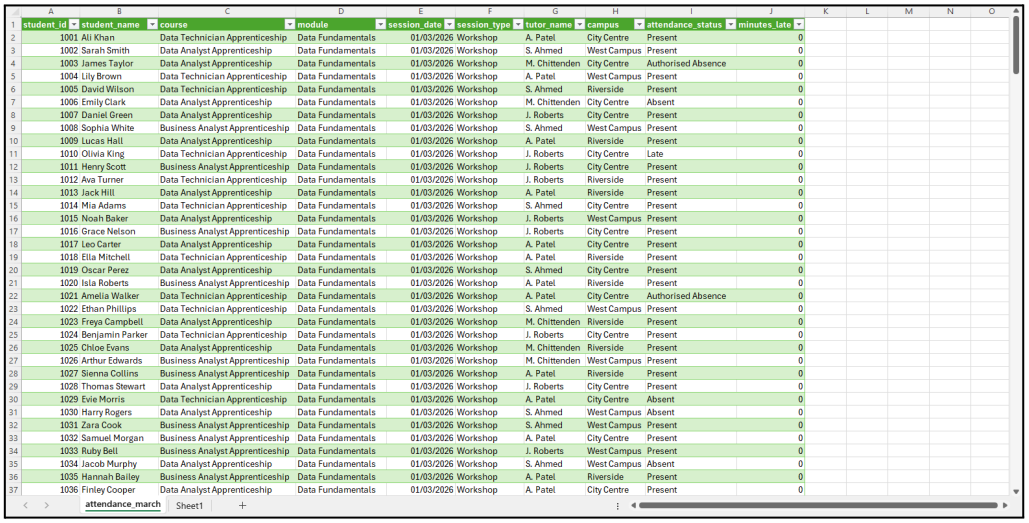
1018	Ella Mitchell	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Pr
1019	Oscar Perez	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	City Centre	Pr
1020	Isla Roberts	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Pr

This will load the table into Excel as a table and part of the data mode;, with data types inferred.



student_id	student_name	course	module	session_date	session_type	tutor_name	campus	attendance_status	minutes_late
1001	Ali Khan	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	City Centre	Present	0
1002	Sarah Smith	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Present	0
1003	James Taylor	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	M. Chittenden	City Centre	Authorised Absence	0
1004	Lily Brown	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	West Campus	Present	0
1005	David Wilson	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	Riverside	Present	0
1006	Emily Clark	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	M. Chittenden	City Centre	Absent	0
1007	Daniel Green	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Present	0
1008	Sophia White	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Present	0
1009	Lucas Hall	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Present	0
1010	Olivia King	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Late	0
1011	Henry Scott	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Present	0
1012	Ava Turner	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	Riverside	Present	0
1013	Jack Hill	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Present	0
1014	Mia Adams	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	City Centre	Present	0
1015	Noah Baker	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	West Campus	Present	0
1016	Grace Nelson	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Present	0
1017	Leo Carter	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	City Centre	Present	0
1018	Ella Mitchell	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Present	0
1019	Oscar Perez	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	City Centre	Present	0
1020	Isla Roberts	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Present	0
1021	Amelia Walker	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	City Centre	Authorised Absence	0
1022	Ethan Phillips	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Present	0
1023	Freya Campbell	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	M. Chittenden	Riverside	Present	0
1024	Benjamin Parker	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Present	0
1025	Chloe Evans	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	M. Chittenden	Riverside	Present	0
1026	Arthur Edwards	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	M. Chittenden	West Campus	Present	0
1027	Sienna Collins	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Present	0
1028	Thomas Stewart	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Present	0
1029	Evie Morris	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	City Centre	Absent	0
1030	Harry Rogers	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Absent	0
1031	Zara Cook	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Present	0
1032	Samuel Morgan	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	City Centre	Present	0
1033	Ruby Bell	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	West Campus	Present	0
1034	Jacob Murphy	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Absent	0
1035	Hannah Bailey	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Present	0
1036	Finley Cooper	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	City Centre	Present	0

attendance_march | Sheet1

What does **Transform Data** do and why don't we use that?

When importing data using **Get Data**, Excel often gives you the option to choose **Transform Data**. This opens the dataset in Power Query so that you can clean or modify it before loading it into Excel. However, it is not always necessary to use this option immediately. At the early stage of working with a new dataset, it is usually helpful to **load the data first and review it in its original form**. This allows you to check that the import has worked correctly, confirm that the columns and rows appear as expected, and understand the structure of the data before making any changes. Once you are confident that the data has imported correctly, you can then return to Power Query later to perform cleaning or transformation tasks. Taking this approach helps avoid making unnecessary changes and ensures that you first understand the dataset you are working with.

Practice Activity

Import the **attendance_march.csv** file.

Once imported:

- Scroll through the table
- Identify the column names
- Check how many rows are present
- Convert the dataset into an **Excel Table**

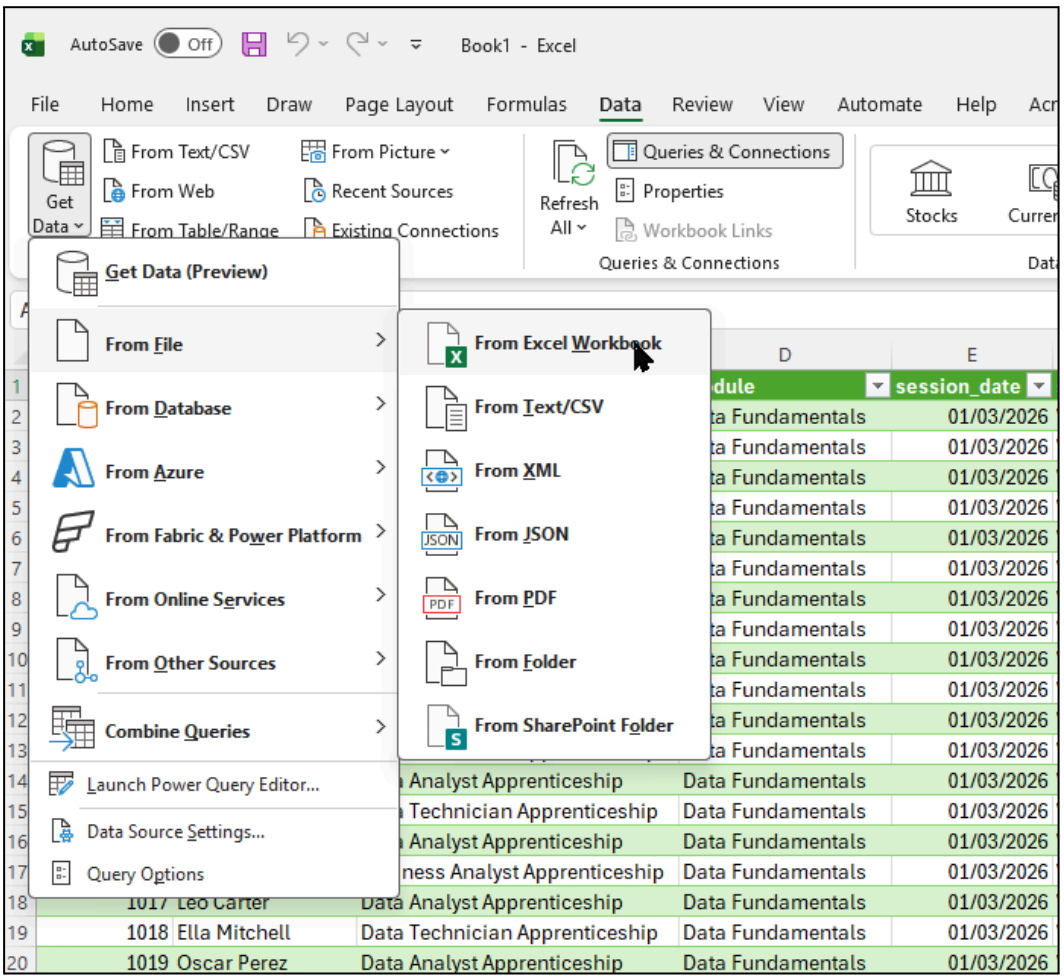
Skill 2 – Importing an Excel Workbook

Data is often stored in another Excel file.

Instead of copying and pasting data manually, Excel can connect directly to the workbook.

Step-by-Step

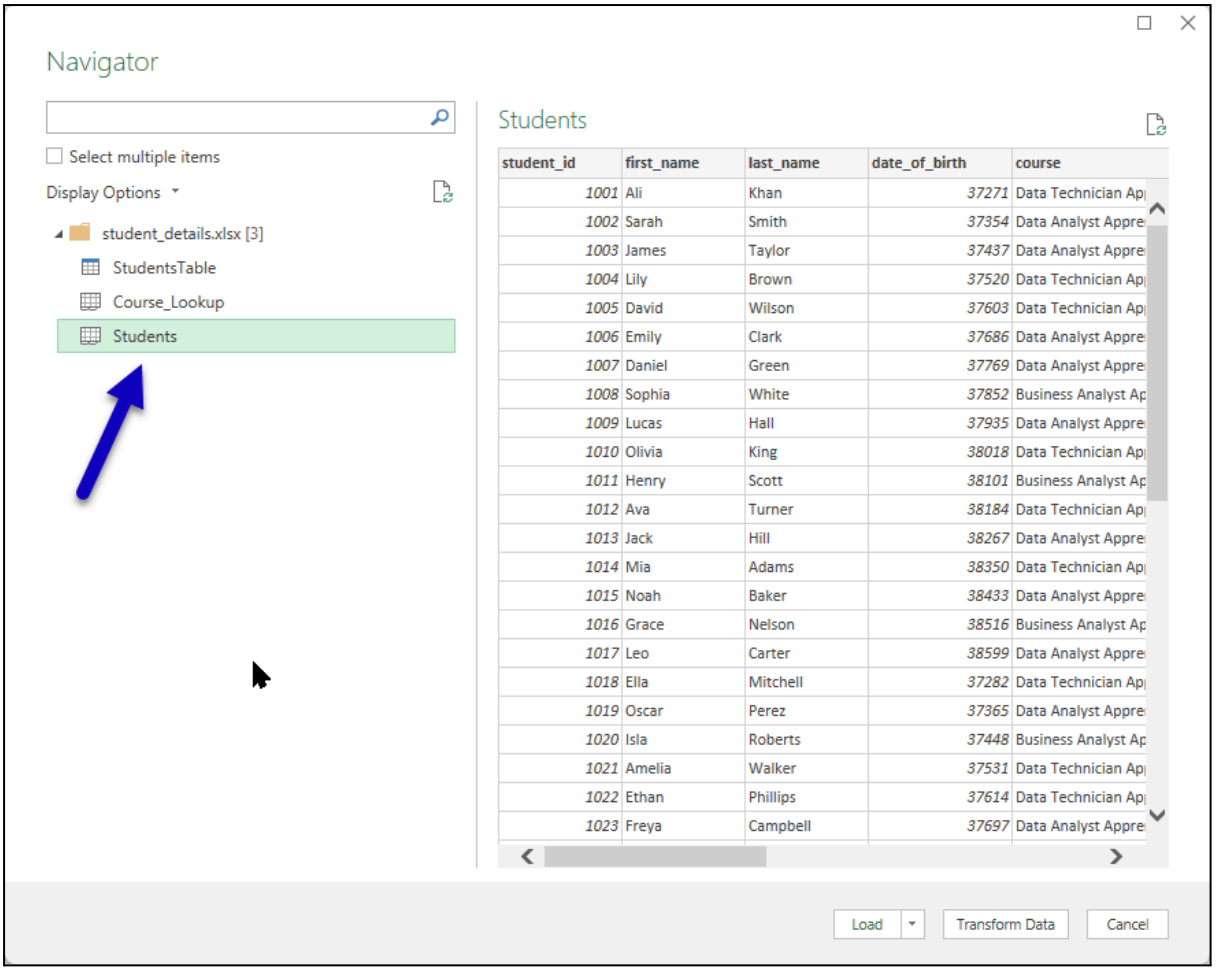
1. Select the **Data** tab
2. Click **Get Data**
3. Choose **From File → From Excel Workbook**



4. Select the file **student_details.xlsx**

Excel will display a **Navigator window** showing available sheets.

Select the sheet called **Students**.



Click **Load**.

Practice Activity

Import the **student_details.xlsx** file.

After importing:

- Review the column names
- Check the student ID column
- Look at how many students are listed

Skill 3 – Importing a Second CSV File

Repeat the CSV import process using the file:

engagement_scores.csv

This file contains student engagement scores.

Practice Activity

Import the file and check:

- Student ID column
- Engagement score column
- Total number of records

Notice that the **student ID field appears in multiple datasets**.

This field is important because it allows data to be linked together later.

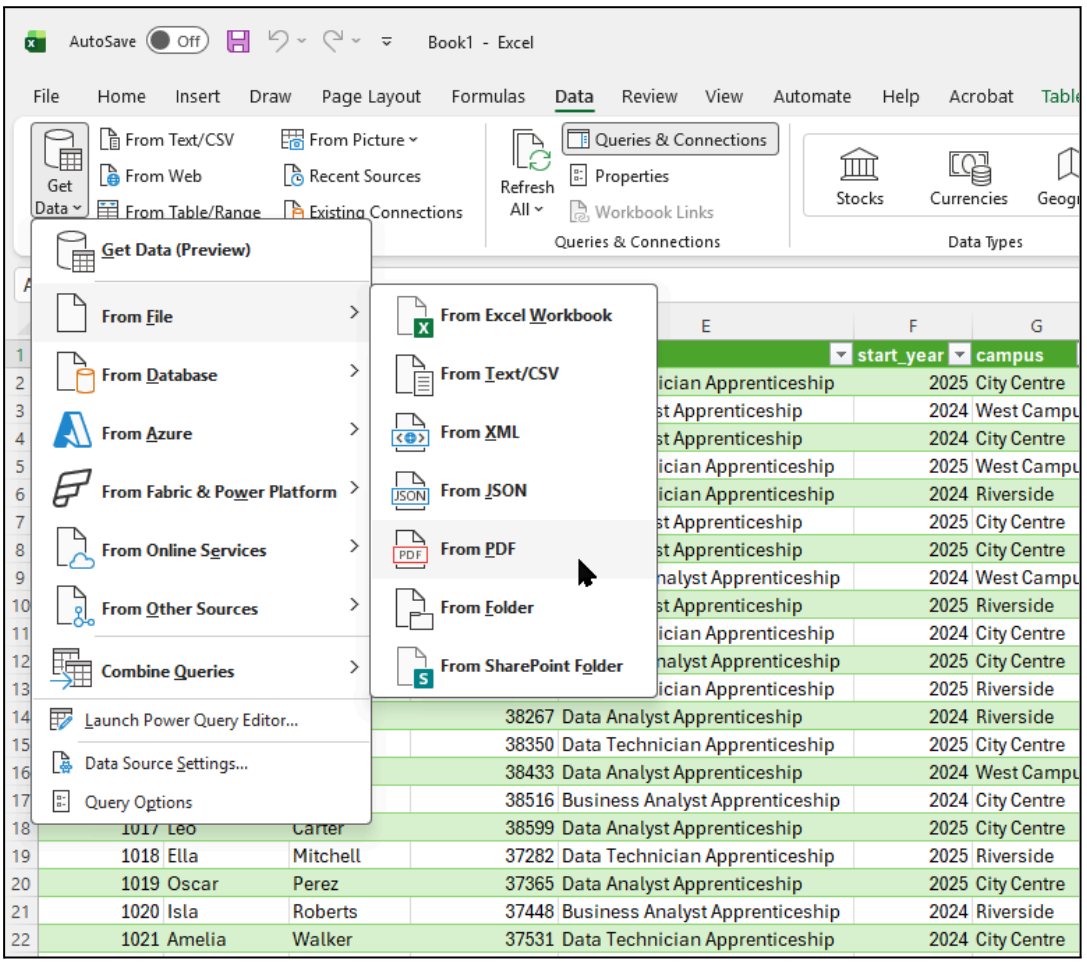
Skill 4 – Importing a Table from a PDF

Some systems export reports as **PDF documents**.

Excel can detect tables inside a PDF and import them directly.

Step-by-Step

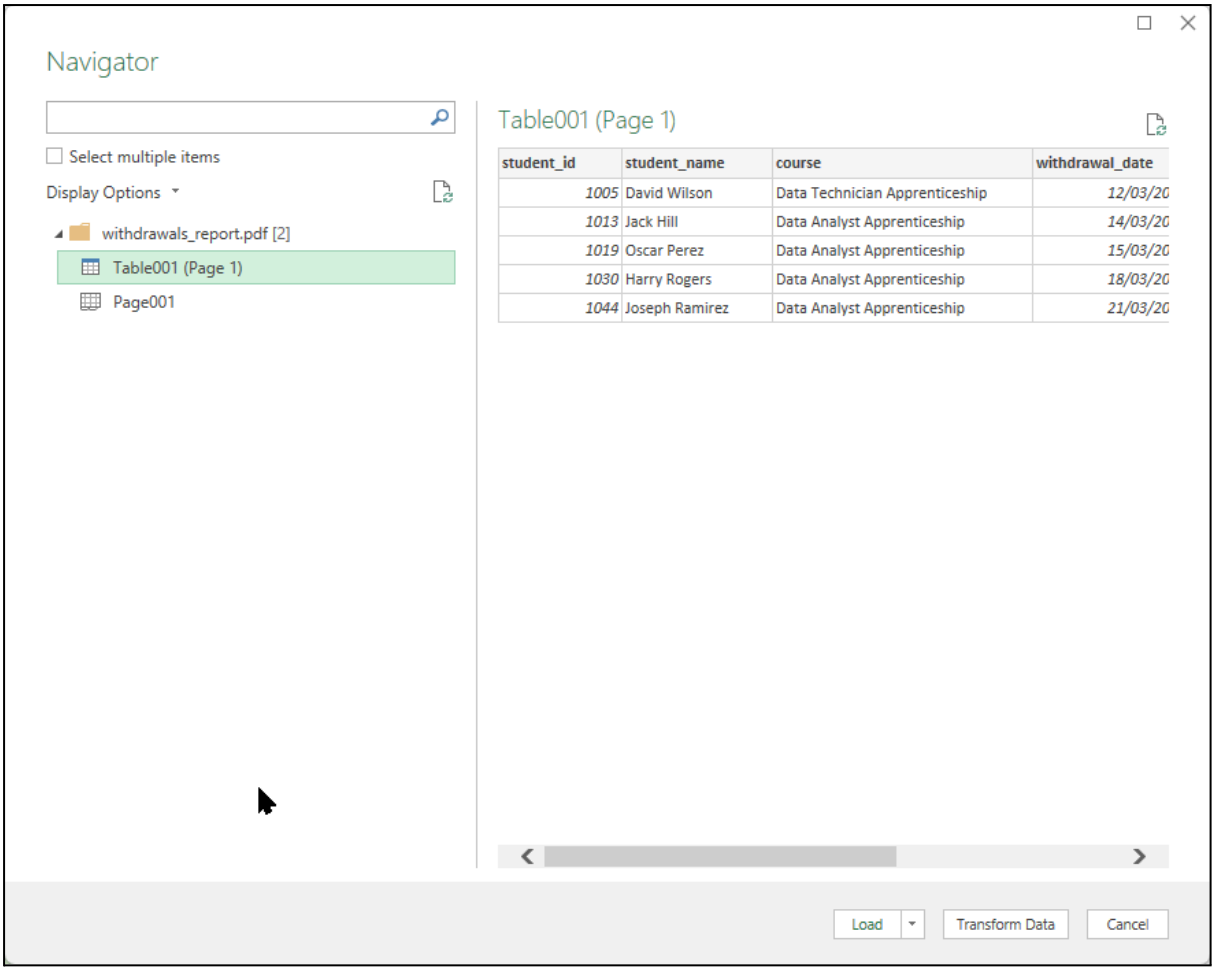
1. Select **Data**
2. Click **Get Data**
3. Choose **From File → From PDF**



4. Select the file **withdrawals_report.pdf**

Excel will scan the document and show any tables it finds.

Select the table that contains the withdrawal information (in this case Table001).



Click **Load**.

Practice Activity

Import the withdrawal report.

After importing:

- Review the columns
- Check the withdrawal date column
- Confirm how many students appear in the report

What You Should Now Have

At the end of this skill builder you should have **four separate tables in Excel**:

- Attendance data
- Student details
- Engagement scores
- Withdrawal records

Each dataset has been imported from a different file type.

This is the **first step in preparing data for analysis**.

Key Concept

A data analyst rarely starts with a clean dataset.

Instead, the first stage is usually:

Import → Prepare → Analyse

This skill builder focused on the **Import stage**.

Later skill builders will cover:

- Cleaning data
- Combining datasets
- Analysing data
- Creating dashboards